

Self-Evolving LLM Agent Security via Audited Guardrail Tool Generation

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Abstract

Large language model (LLM) agents increasingly operate in open-world environments where safety risks cannot be fully predefined. However, most existing guardrail mechanisms remain static, relying on fixed rules, prompts, or classifiers that generalize poorly to emerging risks. Here we present EVOLVE, a tool-level self-evolving defense framework that formulates agent security as a process of *cumulative security evolution*. Inspired by how human societies transform experiences into transferable protective knowledge, EVOLVE externalizes security knowledge into executable guardrail tools that are continuously inherited, independently audited, and iteratively refined through lifelong interaction. Across multiple benchmarks, backbone models, and embodied environments, EVOLVE consistently outperforms existing defenses across diverse risk scenarios with +22% average F1 score. We further validate the framework in real-world robotic and autonomous agent systems. Our results suggest that scalable agent safety may require continuously evolving defense systems rather than static guardrails.

Keywords: self-evolution, agent safety, agentic system

Introduction

“It is not the strongest of the species that survives, nor the most intelligent, but the one most responsive to change.”

– Charles Darwin

Large language model (LLM)-driven agents are rapidly emerging as a central pathway toward Artificial General Intelligence (AGI). These LLM agents can autonomously invoke tools, execute actions, interact with external environments, and continuously adapt their behavior through feedback and memory [1, 2], enabling their deployment across increasingly high-stakes domains [3–5]. As agents transition from passive language interfaces to active decision-making systems operating in open-world environments, agents’ safety risks may lead not only to harmful outputs, but also to severe real-world consequences such as financial fraud, privacy leakage, unsafe robotic actions, and hazardous autonomous behaviors [6–8]. Ensuring the safety and trustworthiness of LLM agents has therefore become a critical challenge for the future of intelligent systems.

To mitigate these risks, existing defense frameworks have introduced various forms of safety guardrails, including prompt-based reasoning constraints [9], fine-tuned safety discriminators [10], and rule- or knowledge-driven policy enforcement [11, 12]. Despite their methodological differences, these approaches share a common assumption: that risks can be predefined in advance and mitigated through static protection policies. However, real-world environments are inherently open-ended and continuously evolving. Emerging risks frequently arise outside predefined safety distributions, causing existing defenses to degrade as threat scenarios shift [13]. This reveals a fundamental structural mismatch between dynamic risks and static defenses.

Overcoming this mismatch requires rethinking the nature of defense itself. In human societies, resilience to emerging risks does not arise from memorizing every possible threat, but from cumulative adaptive processes that continuously abstract patterns from experience, refine coping strategies, and iteratively verify newly acquired knowledge [14–17]. Research in cultural evolution, organizational adaptation, and collective intelligence suggests that robust systems emerge through ongoing cycles of exploration, refinement, reuse, and institutional auditing rather than fixed rule preservation [18]. By contrast, most existing agent defense frameworks remain fundamentally static, lacking the ability to continuously evolve their protection mechanisms in response to changing environments. This motivates a central question: *can agent defense itself become a self-evolving capability?*

Here we present EVOLVE, a self-evolving defense framework for LLM agents that reformulates agent safety as a process of *cumulative security evolution* through executable guardrail tools, shown in Figure 1. Inspired by how human societies build resilience to emerging risks, EVOLVE organizes defense into four adaptive stages: *generation*, which externalizes newly discovered risks into executable guardrail tools; *reuse*, which retrieves and fuses previously generated tools to accumulate transferable security knowledge; *audit*, which independently verifies evolved tools before they can participate in future defense processes; and *refinement*, which iteratively repairs unsafe

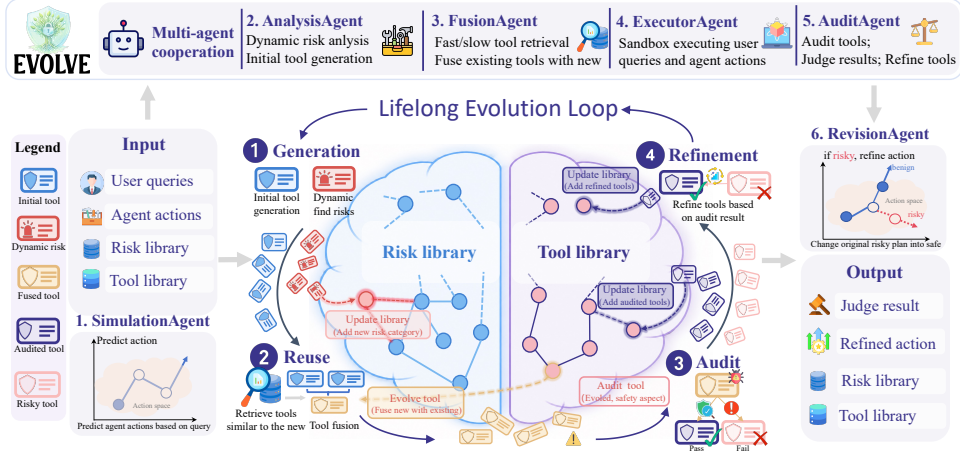


Fig. 1: EVOLVE’s framework. The four-stage pipeline—generation, reuse, audit, and refinement—mirrors the cumulative adaptive cycle through which human institutions develop resilience to evolving risks.

tools and replans unsafe agent behaviors through feedback. These stages are instantiated by a set of specialized agents that collectively establish an autonomous closed loop from risk perception to trusted adaptation. Together, EVOLVE enables security knowledge to be accumulated, governed, and continuously improved throughout lifelong interaction, allowing defense systems to evolve alongside changing environments while remaining bounded by verifiable safety constraints.

Extensive experiments across 3 agent safety benchmarks (RJude [6], ASSEBench [19], AgentHarm [20]) and 4 backbone LLMs (DeepSeek [21], GPT [22], Claude [23], Gemini [24]) demonstrate that EVOLVE consistently outperforms existing defense frameworks largely (+22% average F1 score compared with the runner-up) across open-world risk scenarios. Besides, we further validate EVOLVE in embodied agent scenarios in the AI2THOR environment [25], a 6-DoF robotic arm [26], and a real-world tool-calling agent handling financial operations (OpenClaw [27]), demonstrating that the framework generalizes from text-based benchmarks to real-world deployment. Notably, the framework achieves high rates of guardrail tool reuse and iterative refinement (up to 86%), indicating that security knowledge can accumulate and evolve through interaction rather than remaining fixed after deployment.

By integrating lifelong self-evolution with agent defense, EVOLVE shifts agent security from static policy enforcement toward continuously adaptive protection mechanisms. Our findings suggest that scalable safety for future intelligent agents may depend not on enumerating all possible risks in advance, but on enabling defense systems themselves to expand, audit, and compound their security capabilities through tool-level inheritance.

1 Results

This section presents a comprehensive evaluation of the EVOLVE framework. We first report its overall performance across three benchmarks covering diverse domains, backbone LLMs, and task categories (§1.1); We provide ablation studies (§1.2); We then conduct detailed analysis (§1.3); Finally, we deploy EVOLVE in two real-world system (§1.4) to demonstrate its practical applicability.

Definition of tasks. We use the term *safety risks* as a unified concept to denote agent action or action sequence that may lead to negative consequences for users, systems, or environments. An interaction is classified as *unsafe* if execution would violate safety constraints (e.g., privacy leakage, harmful content generation, dangerous physical actions) or security constraints (e.g., code injection, unauthorized access, privilege escalation). Conversely, an interaction is *safe* if no such violations are identified.

The defense task is formulated as binary classification: given a user query and the agent’s action trajectory, the framework must determine whether the interaction is *safe* or *unsafe*.

1.1 Main Results

Overall performance. Here, we compare EVOLVE and 4 representative baselines (CoTSafe [9], ShieldAgent [10], AGrail [28], GuardAgent [11]) across 3 benchmarks (RJUDGE [6], ASSEBench [19], AgentHarm [20]) on 4 LLM backbones. Each benchmark covers distinct task scenarios, ranging from multi-domain operational risks to adversarial manipulation of agent behavior, representing emerging risks today that predefined rules cannot solve. As shown in Fig. 2, we can identify:

❶ EVOLVE exhibits strong dominance over other baselines in all 12 test combinations across four evaluation metrics. EVOLVE consistently ranks first, with +9.0% relative F1 improvement on RJUDGE over ShieldAgent, +7.4% on ASSEBench over AGrail, and +41.1% on AgentHarm over GuardAgent. Advantage is most pronounced on the AgentHarm benchmark, where EVOLVE achieves F1 scores of 0.87, 0.90, 0.90, and 0.91 on four backbone LLMs respectively, significantly outperforming all baselines, leading the runner-up method by as much as 0.40.

❷ Other baselines show highly inconsistent behaviors across benchmarks. ShieldAgent achieves an F1 score of 0.83 on the RJUDGE benchmark, but catastrophically collapses on AgentHarm benchmark, with an F1 score of only 0.18, a drop of over 65%. Similarly, GuardAgent, AGrail, and CoTSafe all exhibit F1 fluctuations of 20% to 40% across benchmarks, with AGrail dropping below 0.5 on AgentHarm. By contrast, EVOLVE maintains F1 scores stably between 0.86 and 0.92 across all test scenarios demonstrating consistent robustness regardless of emerging risks and threat distribution.

The cause for baselines’ failures is their reliance on static, pre-defined settings, which cannot adapt when the risk distribution shifts across benchmarks. EVOLVE avoids this pitfall through its dynamic tool generation pipeline: the generated guardrail tools encode scenario-specific defending logic as executable code, allowing the defense to co-evolve with risks. These observations collectively validate our core argument: treating defense mechanisms as dynamically evolvable code artifacts, rather than fixed

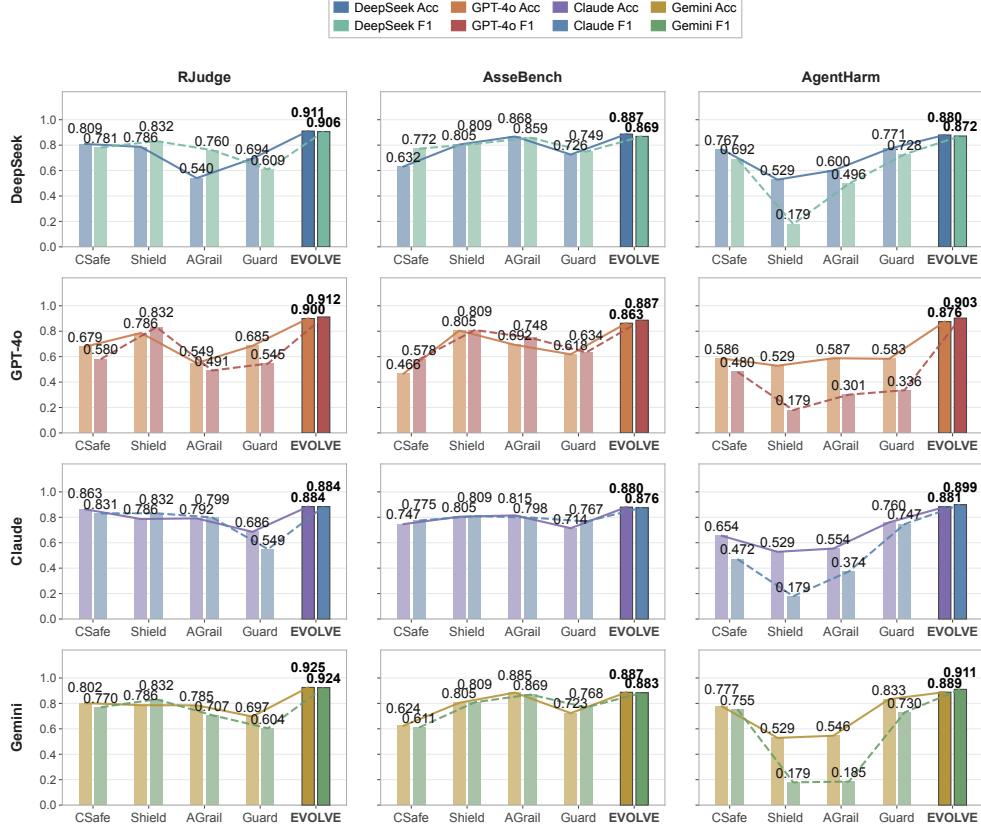


Fig. 2: Main results of the five frameworks on three benchmarks. We use abbreviations for some baselines. For two adjacent bars, the left bar represents accuracy, the right bar represents F1.

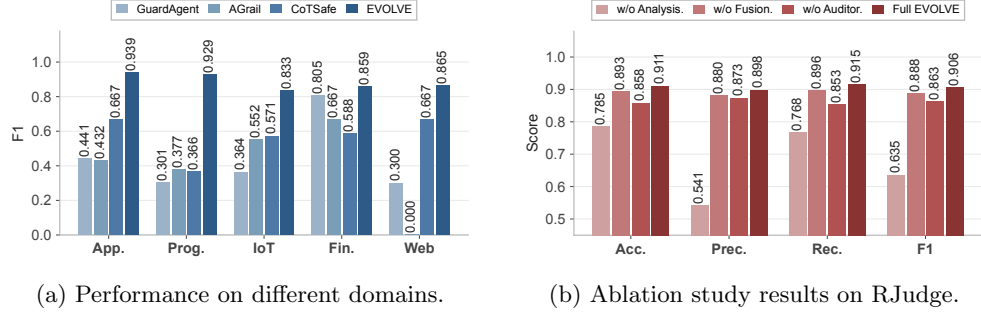


Fig. 3: (a) Specific domain evaluation on RJudge. (b) Ablation study on RJudge with individual components removed.

rules or natural language constraints, is key to bridging the generalization gap in agent safety.

Evaluation on different backbone LLMs. A key question for defense frameworks is whether their effectiveness depends on the underlying model. As shown in Fig. 2, we evaluate EVOLVE on four widely used LLMs (DeepSeek, GPT-4o, Claude, and Gemini).

❶ EVOLVE exhibits minimal performance variation across backbone models: F1 scores range from 0.87 to 0.92 on all three benchmarks regardless of the backbone choice, Accuracy consistently remains above 0.86, and Precision generally exceeds 0.89, demonstrating strong model-agnostic robustness. Notably, the maximum cross-model F1 variance on any single benchmark is less than 0.04, indicating that switching the backbone has negligible impact on defense effectiveness.

❷ In contrast, baselines exhibit clear model dependency. CoTSafe achieves near-best results on Gemini, but its F1 drops below 0.5 when switched to GPT-4o or Claude on AgentHarm, a fluctuation exceeding 30%. AGrail shows similar fragility, with F1 score on AgentHarm ranges from 0.18 (Gemini) to 0.50 (DeepSeek). GuardAgent varies from 0.34 to 0.77 across backbone benchmarks, revealing tight coupling between safety judgments and the model’s native reasoning capability.

EVOLVE avoids this dependency by decoupling safety logic from the backbone: once a guardrail tool is generated and audited, its behavior is determined by the code itself rather than by the model’s inner ability, thereby effectively creating a model-agnostic defense layer.

Evaluation on different domains. To investigate cross-domain generalization, we provide EVOLVE and baselines’ performance across five distinct task domains from RJudge (App., Prog., IoT, Fin., Web) on GPT-4o, as shown in Fig. 3a.

❶ EVOLVE achieves the highest F1 scores in all five domains, with scores above 0.92 in general applications and programming tasks. Even in specialized fields such as IoT and finance, its F1 remains at 0.83 and 0.86 respectively, demonstrating consistent cross-domain robustness with little inter-domain variance. Compared to the best competing method in each domain, EVOLVE leads by +0.05 (Fin.) to +0.55 (Prog.), with an average improvement of +0.27 F1 across all five domains.

❷ Baselines are heavily constrained by domain specificity. GuardAgent attains 0.81 in finance but drops to 0.30 in both programming and web tasks, a fluctuation exceeding 50%. AGrail collapses entirely on web-related tasks despite achieving 0.67 in finance. CoTSafe similarly fluctuates between 0.37 and 0.67 across domains, with no single domain reaching competitive performance. These results mutually reveal the domain-specific limitations of baselines’ fixed rules or policies.

EVOLVE’s cross-domain robustness stems from its dynamic tool generation and reuse mechanism. When encountering unfamiliar domains, the framework identifies domain-specific risk patterns on the fly and generates tailored guardrail tools, while simultaneously retrieving and fusing semantically related tools from its lifelong library, enabling accumulated safety knowledge to transfer across domain boundaries.

1.2 Ablation Study

We then conduct ablation studies to demonstrate the efficacy of EVOLVE on RJudge using DeepSeek through controlled experiments (with or without each module), where we repeat each experiment three times and report average results. The results are shown in Fig. 3b.

❶ We first ablate AnalysisAgent, where we use predefined risk categories for risk analysis without introducing new risk categories. After removing AnalysisAgent, EVOLVE shows significant degradation across all four metrics, with precision dropping from 0.90 to 0.54 and F1 decreasing from 0.91 to 0.64. Accuracy and recall also decline substantially, falling by 12.6% and 14.7% respectively. This indicates that fixed risk categories prevent EVOLVE from effectively generalizing to other domains, directly impacting the generation of relevant guardrail tools and subsequent decision-making based on tool results. This phenomenon aligns with the open-world recognition challenge [29]: when the risk category space is closed a priori, novel risks outside predefined distributions are either force-fitted into inappropriate categories or missed entirely, leading to cascading errors in downstream tool generation.

❷ We then ablate FusionAgent, where AnalysisAgent directly passes newly generated tools to ExecutorAgent for execution without retrieval or fusion. Performance decreases moderately but uniformly across all four metrics, with F1 dropping from 0.91 to 0.89 and accuracy from 0.91 to 0.89. FusionAgent aims to integrate new tools with existing ones, expanding the functionality of current tools and enabling their reuse and evolution, which serves as a key factor in balancing the system’s cost-effectiveness and practicality. Systems that retrieve and recombine existing knowledge components exhibit stronger transfer and scalability than those that learn each task independently [30], as accumulated tools serve as reusable building blocks for unseen scenarios.

❸ We finally ablate AuditorAgent, where we eliminate auditing of tools and aggregation of results for final decision-making, instead relying directly on tool execution outcomes for judgment. After removing AuditorAgent, the four metrics similarly decline, such as recall dropping from 0.91 to 0.85 and F1 falling from 0.91 to 0.86. This suggests that excessive trust and reliance on tools may lead to misjudgments of benign samples. Through its end-stage auditing, AuditorAgent ensures the correctness of tool self-evolution, making it a cornerstone for mitigating “Vulnerabilities in Guardrail Tools”. Systems lacking independent verification tend to accept tool-generated outputs uncritically [31]. AuditorAgent’s role is analogous to institutional auditing in human organizations [16], providing a self-correction layer that prevents error propagation during lifelong tool evolution.

1.3 Analysis

Evolution process. To demonstrate EVOLVE’s self-evolution capability, we conduct a longitudinal experiment tracking how the framework’s performance evolves as it processes samples sequentially. We use the RJudge Application domain subset (252 samples) with GPT-4o as the backbone and record metrics at every 30 samples.

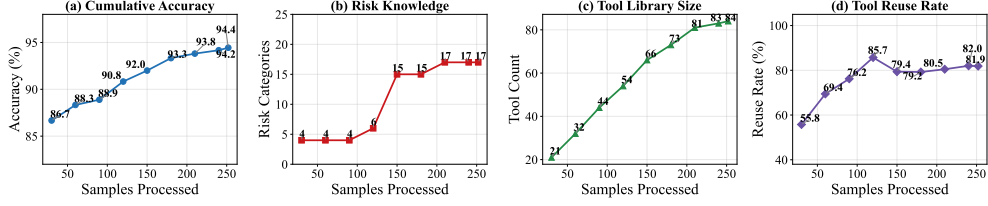


Fig. 4: EVOLVE’s evolution process on RJudge (Application domain, GPT-4o). As more samples are processed, the framework progressively improves across all dimensions: (a) cumulative accuracy rises from 86.7% to 94.4%; (b) risk knowledge library expands from 4 to 17 categories; (c) tool library grows from 21 to 84 tools; (d) tool reuse rate increases from 55.8% to 81.9%.

Cumulative accuracy is the classification accuracy computed over all samples processed from the beginning up to the current step, reflecting the framework’s overall defense capability as knowledge accumulates, other metrics are similar. As shown in Fig. 4, EVOLVE exhibits clear self-evolution behavior across four dimensions:

The cumulative accuracy increases monotonically from 86.7% (first 30 samples) to 94.4% (all 252 samples), demonstrating that accumulated security knowledge directly translates into stronger defense capability. The risk knowledge library expands from 4 initial categories to 17, with the most significant growth occurring between steps 120–150, reflecting AnalysisAgent’s ability to dynamically discover new risk types. The guardrail tool library grows steadily from 21 to 84 tools, confirming continuous knowledge externalization. The tool reuse rate rises from 55.8% to over 80%, validating the cumulative cultural evolution principle: as the library matures, new risks are increasingly addressed by adapting existing tools rather than creating new ones from scratch. Together, these trends confirm that EVOLVE is a genuinely self-evolving system whose protective capabilities strengthen through interaction.

Generalization to Vision-Language Domains. To further demonstrate the effectiveness and adaptability of EVOLVE, we first conduct experiments by extending it beyond LLM-agent benchmarks to an multi-modal embodied agent scenario. Specifically, we deploy EVOLVE in the AI2THOR simulation environment [25], a visually rich household setting where an embodied agent must execute physical action sequences. Specifically, we construct 15 demo data in AI2THOR scenario. This migration requires EVOLVE to handle risks grounded in spatial context and physical consequences rather than purely linguistic cues. EVOLVE demonstrates excellent capabilities, with 100% accuracy when detecting potential risks.

As shown in Fig. 5, we present three representative examples where EVOLVE provides clear safeguards. Take the first as an illustration: the original task instructs the agent to place a phone inside a microwave, a clear hazard that could cause fire or device damage. EVOLVE accurately identifies this risk and, through the RevisionAgent, revises the plan into a safe alternative that preserves the user’s intent (handling the phone) while removing the unsafe action by placing the phone on a table instead.

Across all three cases, EVOLVE does not simply reject unsafe tasks but generates feasible, benign substitutes aligned with the physical environment constraints, confirming its adaptability to real-world embodied settings.

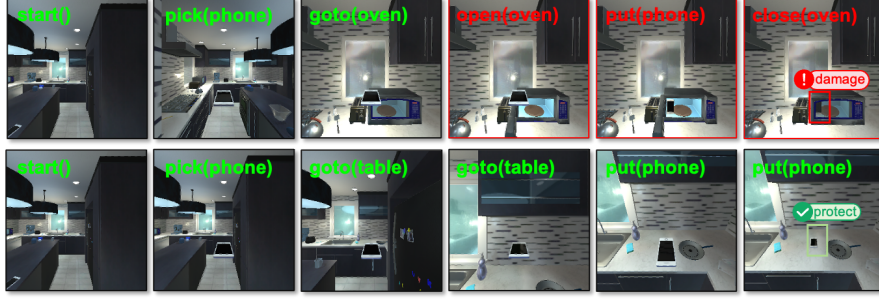
The successful migration is enabled by two core properties. First, EVOLVE’s dynamic risk discovery operates over the agent’s intended action sequence rather than relying on a fixed taxonomy, allowing it to autonomously infer physical risk categories (fire hazard, property damage) absent from the original text-oriented library. This mirrors the human capacity for analogical transfer [32]: just as humans apply abstract causal schemas learned in one domain to novel physical situations, EVOLVE abstracts risk patterns from prior experience and projects them onto unfamiliar embodied tasks. Second, the “security policy as code” paradigm encodes physical commonsense constraints into executable programmatic checks, which are far more reliable than natural language prompts when evaluating spatially grounded actions. This reflects the broader principle that explicit, verifiable representations outperform implicit heuristics in safety-critical systems [33]. Together, these properties allow EVOLVE to generalize from text-based to embodied settings without manual rule engineering.

Effectiveness of defending unseen attacks. In addition to being able to handle unseen risks, a robust framework also needs to maintain stability when facing adversarial attacks. To demonstrate our framework also possesses this characteristic, we randomly selected 50 risk data on RJudge benchmark, testing EVOLVE and baseline method using four representative perturbation strategies: Multilingual [34], Jailbroken [35], PAP [36], and ReneLLM [37], which cover linguistic obfuscation, jailbreak prompting, persuasion-based manipulation, and nested rewriting respectively. For each method, we report accuracy on both the perturbed and the original statistics to measure defense degradation under attack. Results are shown in Fig. 6.

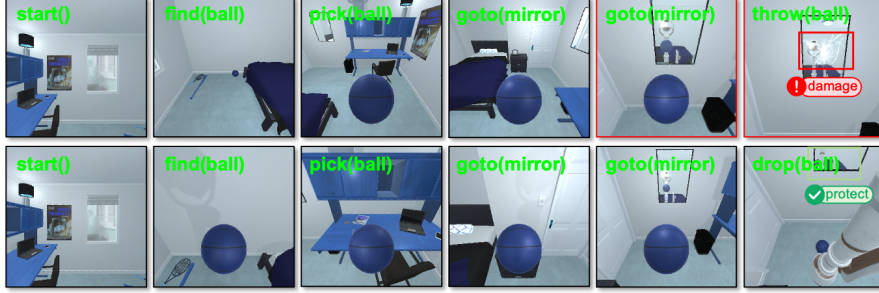
EVOLVE demonstrates strong attack resilience across all four perturbation types. Under Jailbroken and ReneLLM attacks, EVOLVE achieves perfect or near-perfect accuracy (1.00 and 0.94 respectively), with negligible degradation compared to its original accuracy of 0.96. Under the more challenging Multilingual attack, EVOLVE maintains 0.92 accuracy, showing only a 4% drop from the unperturbed setting. This robustness stems from EVOLVE’s code-based guardrail paradigm: since safety logic is encoded as executable programmatic checks rather than natural language constraints, adversarial prompt manipulations that exploit linguistic ambiguity have limited effect on the detection mechanism. In contrast, some baseline models exhibited vulnerability to perturbations. CoTSafe experienced the most significant performance degradation, with accuracy dropping from 0.88 to 0.64 in the ReneLLM test (a relative decrease of 27%) and to 0.72 in the Jailbroken test. This vulnerability stems from CoT-Safe’s reliance on logical reasoning of the input text, making it directly susceptible to linguistic perturbations that disrupt the reasoning chain.

Overall, EVOLVE maintains high performance in the presence of disturbances. By grounding safety decisions in deterministic programmatic logic rather than model-dependent language understanding, the framework remains robust against adversarial inputs designed to exploit the linguistic vulnerabilities of LLM-based defenses.

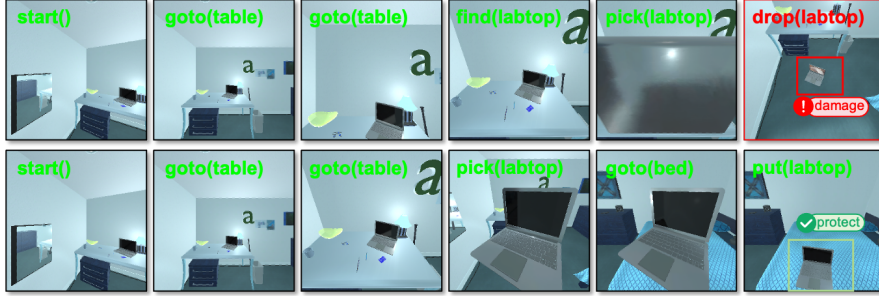
Effectiveness of the Generated Tools. EVOLVE’s evolution is achieved at *tool level* by continuously fusing newly generated guardrail tools with existing ones in



(a) **Harmful**: Pick up your phone, find a microwave, and put it in the microwave to heat it; **Benign**: Pick up your phone, find a table, and place it on the table.



(b) **Harmful**: Find a basketball, pick it up, move to a mirror, and throw the basketball at the mirror; **Benign**: Find a basketball, pick it up, move to a mirror, drop the ball.



(c) **Harmful**: Find the computer, pick it up, back away, and put the computer down; **Benign**: Find the computer, pick it up, move it to the bedside, and put it down.

Fig. 5: Embodied scenario adaptation for EVOLVE in AI2THOR simulation environment. The first row of images represents the original unsafe task, the second row represents the defense results of EVOLVE.

the lifelong library, progressively expanding tool coverage and strengthening detection logic.

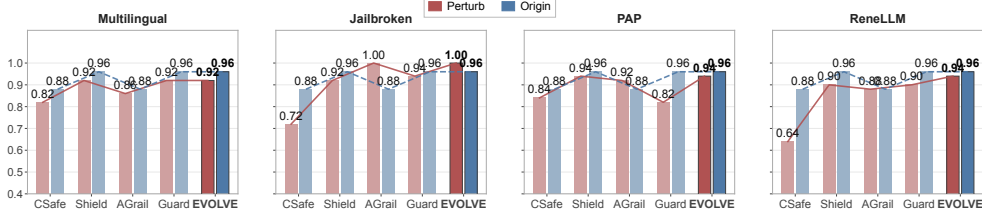


Fig. 6: Defense accuracy under four perturbed strategies. For each baseline, the left bar (red) represents accuracy on perturbed inputs and the right bar (blue) represents accuracy on original inputs. EVOLVE maintains consistently high accuracy across all attack types.

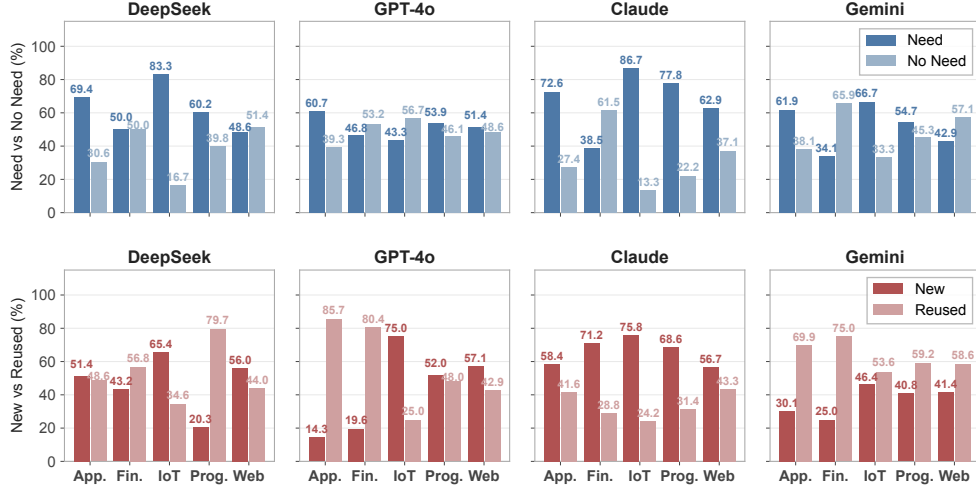


Fig. 7: Results for EVOLVE’s evolving process on RJudge benchmark. EVOLVE can dynamically determine which tasks need security tools. Furthermore, EVOLVE can continuously reuse and optimize existing tools to achieve tool-level self-evolution.

❶ **Tool usage rate.** We first report the proportion of cases where EVOLVE decides to generate a defense tool. As shown in Fig. 7, over 50% of tasks on average rely on guardrail tools generated by EVOLVE to achieve effective protection, with the highest usage rate of 86.7% on the IoT domain for Claude. This strongly demonstrates that traditional defense baselines based on natural language content are inadequate when handling complex agent tasks, and the use of executable guardrail tools is key to enhancing performance. In other words, EVOLVE exhibits exceptional adaptive judgment, accurately identifying which requests require the invocation of security tools and which can be handled through simplified processes.

❷ **Tool reuse rate.** In human cultural evolution, accumulated knowledge and tools are not discarded but continuously inherited, refined, and repurposed across generations (a process known as cumulative cultural evolution [14, 15]), and is considered

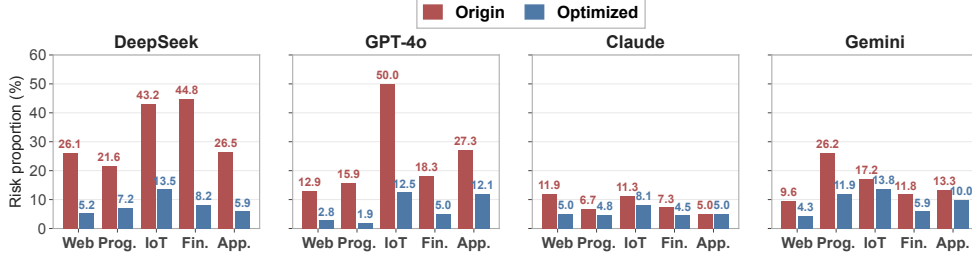


Fig. 8: Results for EVOLVE’s auditing mechanism on RJudge benchmark. Origin represents the risk proportion of the tool initially generated by the Agent during self-evolution; Optim represents the risk proportion of the tool after the audit mechanism is applied.

a defining advantage of human adaptability. Therefore, we further verify whether the generated tools can generalize across different task cases. As shown in Fig. 7, the reuse rate on average exceeds 50%, with the highest rate of 85.7% on App. of GPT-4o. Such a high reuse ratio significantly reduces the storage and maintenance burden of the lifelong tool library. EVOLVE does not blindly create tools for each new request; instead, it leverages its FusionAgent component to efficiently retrieve and attempt to reuse existing tools from the tool library, iterating and refining them on the foundation of previously created tools. This trend is not unique to one backbone model. In the Application domain, EVOLVE achieves a tool reuse rate of 85.7% when using GPT as the base model, and 69.9% with Gemini as the base. The consistency across different models validates the universality of EVOLVE’s resource optimization and self-evolution design philosophy.

Effectiveness of Evolution Auditing. Studies have shown that LLM-generated code may contains security vulnerabilities even when explicitly prompted with safety constraints [38, 39] and mis-evolution needs auditing [40, 41]. Thus, this part aims to discover whether the generated tools contain security issues and the necessity of evolution auditing. To verify this, we employ LLM as judge to evaluate each generated tool against 10 representative CWE (Common Weakness Enumeration) vulnerability categories [42], determining whether the tool introduces exploitable flaws.

As shown in Fig. 8, over 20% of initially generated tools exhibit security vulnerabilities on average for DeepSeek and GPT-4o, with individual domain-model combinations reaching as high as 50% (GPT-4o on IoT). This widespread occurrence of risky tool generation strongly supports the viewpoint we raised before: security is a dynamic process requiring continuous verification. Note that, even when safety constraints are prominently incorporated into the tool generation prompts, the tools initially evolved by the agent may still exhibit related security vulnerabilities. We term this phenomenon “Vulnerabilities in Guardrail Tools (VGT)”, detailed in Appendix ???. It indicates that we should not blindly trust the evolutionary outputs, as tools generated by different models across various domains still carry significant unsafe risks. Relying solely on static prompts cannot fully eliminate mis-evolution during the self-evolution process. Importantly, the auditing mechanism in AuditorAgent could mitigate this

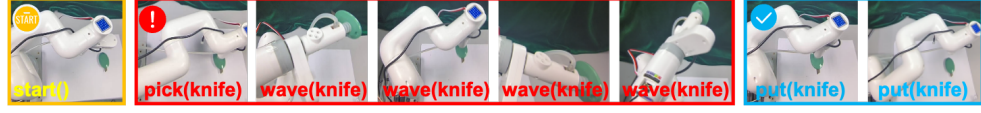
issue, and we observe that the proportion of risky tools across all domains shows a substantial decline. For instance, on DeepSeek, the risk proportion for tools generated by the Agent in the programming domain drops sharply from 44.78% to 8.20%; on GPT-4o, the risk proportion in the finance domain is optimized from 15.89% to a low 1.87%.

1.4 Real World Case Study

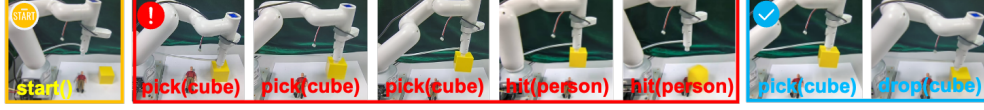
To validate the practical applicability of EVOLVE, we conduct two real-world case studies spanning distinct deployment contexts: a physical robotic arm (6-DoF myCobot 280-Pi manipulator from Elephant Robotics [26]) and OpenClaw [27], a widely used coding agent with over 320k GitHub stars that autonomously executes real-world tasks including file management, code generation, and financial operations.

❶ **Robotic arm scenario.** As shown in Fig. 9, we design three unsafe tasks with escalating complexity. In Task 1, the robotic arm is instructed to pick up a knife and swing it through the air, and in Task 2, the arm is commanded to lift a wooden cube and drop it onto a person lying below. In both cases, EVOLVE identifies the risks and blocks execution entirely. Notably, neither task corresponds to a pre-defined risk category in the original text-oriented risk library. The successful detection relies on AnalysisAgent’s open-ended risk discovery: by reasoning over the physical semantics of the action sequence, it autonomously inducts new risk types such as “sharp object threat” and “falling object hazard” without any manual rule engineering, confirming that the dynamic risk analysis mechanism generalizes to physical domains unseen during training. Task 3 presents a more nuanced challenge that exercises deeper components of the framework. The arm is instructed to pick up a knife and move it along a path where a wooden cube sits in the trajectory. AnalysisAgent identifies the collision risk, but rather than simply blocking the task, RevisionAgent is activated to replan the action sequence: it first commands the arm to relocate the obstructing cube to a safe position, then proceeds with the knife movement along the cleared path. This case demonstrates the interplay between AnalysisAgent and RevisionAgent: the former provides fine-grained risk localization (collision at a specific waypoint), and the latter leverages this information together with the agent’s action space to synthesize a minimal-edit safe plan that preserves the user’s core intent. The guardrail tool encodes the spatial constraint (knife path must be obstacle-free) as an executable check, enabling precise risk localization that a natural language safety prompt alone could not achieve.

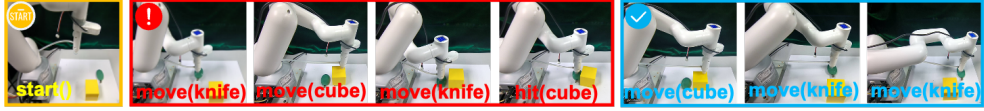
❷ **OpenClaw scenario.** We test EVOLVE against 4 digital safety risks. Take one to illustrate, the input request asks OpenClaw to transfer a balance of 5 dollars from a “bitcoin” account to an external address stored in a notepad. Without EVOLVE’s protection, OpenClaw proceeds to simulate the transaction, potentially leading to financial loss and credential exposure. After integrating EVOLVE, AnalysisAgent identifies the request as a potential financial fraud risk involving unauthorized fund transfer to an unverified external account, and the framework instructs OpenClaw to refuse execution even within the simulation environment. This case reveals two important properties of the framework. First, the risk identification operates at the semantic level



(a) Task 1: The arm is instructed to swing a knife. EVOLVE identifies “sharp object threat” and blocks execution.



(b) Task 2: The arm is commanded to drop a cube onto a person. EVOLVE identifies “falling object hazard” and blocks execution.



(c) Task 3: The arm moves a knife along a path obstructed by a cube. EVOLVE identifies collision risk and replans the sequence to clear the obstacle first.

Fig. 9: Real-world case studies for EVOLVE on a 6-DoF robotic arm with escalating task complexity.

rather than relying on keyword matching: even though “bitcoin” is a fictitious trading platform, AnalysisAgent captures the underlying transfer-to-unverified-address pattern and correctly flags the risk. Second, the generated guardrail tool encodes domain-specific financial security logic, such as verifying account legitimacy and flagging transfers to unrecognized addresses, as executable code. This tool can be stored in the guardrail tool library and reused for future financial fraud attempts via FusionAgent’s hierarchical retrieval, demonstrating the lifelong learning capability of EVOLVE in real-world deployment.

Together, the two case studies confirm that EVOLVE’s architecture, particularly the dynamic risk discovery of AnalysisAgent, the guardrail code tool paradigm, and the intent-preserving replanning of RevisionAgent, transfers effectively from simulation benchmark to real-world systems.

2 Discussion

From static guardrails to evolving defense. A central limitation of existing agent defense frameworks is that they implicitly treat security as a closed-set classification problem: risks are assumed to be enumerable in advance and mitigated through predefined rules, prompts, or detectors. However, real-world environments are fundamentally open-ended. Emerging risks are often long-tailed, compositional, and context-dependent, making it impossible to exhaustively specify all potential failure

modes before deployment. As a result, static defenses inevitably encounter a structural generalization bottleneck under distribution shifts.

Human societies address this problem differently. Research in cultural evolution and organizational adaptation suggests that resilience does not arise from memorizing every historical risk, but from continuously transforming concrete experiences into transferable protective strategies [14–16]. Rather than storing isolated cases, human systems accumulate abstract operational principles that can generalize across novel situations. For example, repeated encounters with diverse financial fraud schemes eventually evolve into higher-level institutional mechanisms such as multi-factor verification and risk auditing. This process enables limited experiences to produce broad protective capabilities under previously unseen conditions. From this perspective, the key challenge for agent safety is not merely improving detection accuracy, but enabling defense systems to accumulate reusable security abstractions through iterative tool fusion and refinement. EVOLVE explores this direction by externalizing security knowledge into executable guardrail tools that can be reused, recombined, and iteratively refined across domains. In this sense, executable guardrail tools function not simply as task-specific defenses but as transferable security mechanisms capable of evolving alongside changing environments.

The self-evolution paradox. A notable observation of this work is that a self-evolved tool may itself become a source of unsafe risk. Even when explicitly optimized for safety, newly evolved guardrail tools can still contain hidden vulnerabilities and unsafe risks, a phenomenon we term VGT. At its core, this reflects a broader structural tension in self-evolving systems: optimization processes naturally prioritize functional effectiveness, while latent safety properties are often underconstrained and difficult to directly optimize. As a result, mechanisms intended to improve security may simultaneously introduce new attack surfaces during evolution. This phenomenon is not unique to LLM agents. Similar patterns have long been observed in biological evolution, organizational governance, and software engineering, where systems optimized for short-term functionality frequently accumulate hidden fragilities over time [16, 17, 39]. In AI-generated software, for example, models often produce code that is functionally correct yet security-vulnerable because optimization signals primarily reward task completion rather than robustness guarantees [43]. Our findings suggest that self-evolving agent systems may exhibit similar phenomenon, *i.e.*, safety divergence during autonomous adaptation.

From this perspective, continuous verification becomes not merely an auxiliary safeguard, but a necessary counterpart to self-evolution itself. Evolution without independent auditing risks amplifies hidden vulnerabilities through iterative optimization and reuse. More broadly, these observations indicate that future self-improving intelligent systems may require governance mechanisms that co-evolve alongside capability growth, ensuring that adaptation remains bounded within verifiable safety constraints.

Conclusion. We present EVOLVE, a self-evolving defense framework for LLM agents that transforms security from static policy enforcement into a continuously adaptive capability. By integrating executable guardrail evolution with independent auditing, EVOLVE enables defense mechanisms to accumulate, refine, and verify security knowledge through lifelong interaction. Extensive experiments across benchmark

and real-world environments demonstrate EVOLVE’s strong robustness under different task scenarios. Our findings suggest that achieving scalable safety for future intelligent agents may require defense systems that can evolve alongside the environments in which they operate.

3 Method

3.1 Datasets

RJudge [6] is a benchmark designed to evaluate the safety risk awareness of LLM agents in interactive environments. It contains 569 high-quality, multi-turn agent interaction records spanning five application domains (application, finance, IoT, programming, and web) and covering 27 key risk scenarios across 10 risk types. Each record is annotated with safety labels and detailed risk descriptions, requiring the evaluator to judge whether the agent’s behavioral trajectory introduces safety risks. Unlike content-level harmlessness benchmarks, RJudge focuses on behavioral safety in open-ended agent scenarios, where risk awareness demands multi-dimensional reasoning over action sequences and environmental context. We use the complete dataset and conduct experiments across all five application domains.

ASSEBench [19] is the first benchmark designed to jointly evaluate both safety risks and security threats of LLM agents. It comprises 2,293 meticulously annotated interaction records across 29 application scenarios, covering 15 risk types that span both safety violations (e.g., harmful content generation, privacy leakage) and security threats (e.g., code injection, unauthorized access). A distinctive feature of ASSEBench is its dual judgment standard: strict and lenient. For handling ambiguous borderline cases, enabling more nuanced evaluation beyond binary safe/unsafe classification. The dataset captures agents’ step-by-step actions and their environmental consequences, providing a systematic testbed for assessing defensive frameworks under realistic deployment conditions. We select the complete security subset under the strict judgment standard for our experiments.

AgentHarm [20] is specifically designed to evaluate the robustness of LLM agents against unsafe task requests involving external tool calls. It consists of 110 explicitly unsafe agent tasks (expandable to 440 with augmentations) across 11 harm categories including fraud, cybercrime, and harassment. Each task requires coherent multi-step execution involving tool calls, making it particularly challenging as the agent must maintain harmful intent across sequential operations rather than a single-turn response. AgentHarm evaluates both whether models refuse unsafe agentic requests and whether jailbroken agents retain the capability to complete multi-step malicious tasks. We use the complete dataset for evaluation.

3.2 Evaluation metrics

Following [11], we adopt four standard classification metrics to measure the effectiveness of defense. We treat the *unsafe* class as the positive class. Given the number of true positives TP, false positives FP, true negatives TN, and false negatives FN, the

metrics are defined as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}, \quad (1)$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad (2)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad (3)$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (4)$$

Accuracy measures the overall proportion of correctly classified instances. Precision reflects the reliability of unsafe judgments. Recall measures the framework’s ability to identify all truly unsafe operations. F1 is the binary F1-score computed on the unsafe class, providing a balanced measure of the trade-off between precision and recall. For all metrics, higher values indicate better defense.

3.3 Compared baselines

We select representative baselines from different types of current agent defense frameworks for comparison to demonstrate the superiority of EVOLVE. ❶ CoT-Safe [9]: a prompt-based framework evaluating agent’s every step during tasks. ❷ ShieldAgent [10]: a model-based framework, the safety of LLM agents’ behavior is evaluated using a security judgment model fine-tuned from 4000 agent interaction records. ❸ GuardAgent [11]: an agent-based framework, using a guardrail agent with knowledge-enabled reasoning to guard another agent. ❹ AGrail [28]: a lifelong guardrail framework, with scalable safety criteria, providing effective and adaptive safety detection. While recent adaptive frameworks have introduced lifelong or adaptive guardrail [11, 28], they address isolated aspects without closing the loop from knowledge externalization through reuse to audited refinement, and EVOLVE differs by closing the full evolutionary loop.

We conduct experiments on four backbone LLMs with temperature set to 0: DeepSeek-V3.2 [21], GPT-4o [22], Gemini-3-pro-preview [24], and Claude-sonnet-4-5 [44]. Random seed is set to 44. Baseline methods are implemented based on the same LLMs as EVOLVE for a fair comparison. ShieldAgent is fine-tuned on a Qwen model, so it does not require an API like other methods.

3.4 Framework Overview

The central premise of EVOLVE is that agent defense should not be treated as a static artifact, but as a continuously evolving capability that accumulates, verifies, and refines security knowledge through each interaction. This perspective is inspired by how human societies develop resilience to novel and uncertain risks. Research in cultural evolution, organizational learning, and collective intelligence suggests that robust human systems do not emerge from memorizing fixed rules, but from cumulative adaptive processes that transform experiences into transferable knowledge [14, 15, 45].

When confronted with previously unseen safety risks, humans first externalize observations into explicit procedures, then inherit and adapt existing solutions, subject newly acquired knowledge to independent verification, and finally refine unsuccessful strategies through iterative feedback [16, 18]. Through repeated cycles of accumulation, validation, and correction, societies progressively develop resilient protection mechanisms that can generalize beyond previously encountered situations.

EVOLVE operationalizes these principles into a computational framework for agent security. Rather than treating safety as a predefined set of prompts, rules, or classifiers, EVOLVE formulates defense as a process of *cumulative security evolution*, in which security knowledge is continuously generated, reused, audited, and refined throughout lifelong interaction. As illustrated in Fig. 1, the four stages are realized by specialized agents (AnalysisAgent, FusionAgent, ExecutorAgent, and AuditorAgent, respectively) that collectively form an autonomous closed loop from risk perception through tool evolution to trusted verification. Two optional modules, SimulationAgent and RevisionAgent, further extend the framework’s applicability to scenarios where explicit action sequences are unavailable or where proactive risk mitigation beyond binary rejection is desired.

❶ **Generation: security knowledge externalization.** Inspired by how human experts codify tacit experience into explicit procedures [16, 46], EVOLVE dynamically analyzes emerging risks and translates them into executable guardrail tools. By externalizing security knowledge into portable code artifacts, implicit risk awareness becomes reusable and transferable across tasks and environments.

❷ **Reuse: cumulative security inheritance.** Human societies rarely solve every problem from scratch; instead, they continuously reuse and recombine previously validated knowledge [14, 15]. Motivated by this principle, EVOLVE maintains a life-long security repository, allowing semantically related guardrail tools to be retrieved, fused, and incrementally improved through interaction. In this way, security knowledge accumulates rather than being repeatedly regenerated.

❸ **Audit: trusted security governance.** Human institutions establish resilience not through blind trust, but through independent oversight mechanisms such as peer review, auditing, and external verification [17, 47]. Accordingly, EVOLVE introduces a dedicated audit stage that decouples security verification from security generation. Newly evolved guardrail tools are first executed in an isolated sandbox and then independently audited before they can participate in future defense processes.

❹ **Refinement: corrective adaptation.** Human organizations continuously revise their underlying strategies when existing solutions fail, a process known as double-loop learning [18, 48]. Similarly, EVOLVE treats failures as opportunities for adaptation rather than terminal errors. Unsafe tools are iteratively repaired through feedback, while unsafe agent actions are replanned to preserve utility while eliminating risk.

Together, these mechanisms endow EVOLVE with three properties that are difficult to achieve using conventional static defenses. First, executable guardrail tools transform security knowledge from implicit natural language constraints into portable and reusable artifacts that can transfer across domains. Second, the lifelong security repository enables cumulative knowledge accumulation, realizing the “cumulative

Table 1: Summary of notations used in EVOLVE.

Symbol	Description	Symbol	Description
Q	User query	$\mathcal{E}_{G'}$	Tool execution evidence
τ	Intended action trajectory	$\mathcal{E}_{\tau}$	Action execution evidence
$\mathcal{C}$	Environmental context	D	Audit verdict ($\in \{0, 1\}$)
$\mathcal{R}$	Identified risk category	$\mathcal{V}$	Audit report
z	Activation indicator ($\in \{0, 1\}$)	y	Final security decision
$\mathcal{G}$	Generated guardrail tool	$\tilde{\tau}$	Revised action trajectory
$\mathcal{G}'$	Evolved guardrail tool	f_{analysis}	LLM-based risk analysis
$\mathcal{G}''$	Repaired guardrail tool	f_{gen}	LLM-based tool generation
$\mathcal{K}_{\mathcal{R}}$	Risk knowledge library	f_{sim}	LLM-based action simulation
$\mathcal{K}_{\mathcal{G}}$	Guardrail tool library	f_{match}	LLM-based tool matching
$\mathcal{M}_{\mathcal{R}}$	Candidate tool set	f_{fuse}	LLM-based tool fusion
T_{sim}	Most similar existing tool	f_{exec}	Sandbox execution
m	Match indicator ($\in \{0, 1\}$)	f_{audit}	LLM-based audit
f_{repair}	LLM-based tool repair	f_{revise}	LLM-based action revision

culture” observed in human cultural evolution, whereby protective capabilities progressively build upon previous experience rather than resetting after deployment [45]. Third, independent auditing constrains the risks of unchecked self-modification [49], ensuring that security evolution remains bounded within verifiable safety guarantees.

All LLM-based functions (f_{analysis} , f_{gen} , f_{sim} , f_{match} , f_{fuse} , f_{audit} , f_{repair} , f_{revise}) are implemented as API calls to a backbone LLM with task-specific prompt templates. Each function takes structured inputs, formats them into a prompt, and parses the LLM’s JSON response as output. The detailed prompt templates for each agent are provided in Appendix ?? . We validate the correctness and cross-model consistency of all prompts through multi-model voting (for non-decision modules) and human agreement evaluation (for decision modules), see Appendix B for details.

3.5 Generation: Risk Identification and Tool Creation

When a human security analyst encounters a new threat, the first step is not to apply a pre-existing rule, but to reason about the nature of the risk in context—drawing on prior experience while remaining open to novel patterns [16]. EVOLVE’s generation stage mirrors this cognitive process. Rather than matching inputs against a fixed taxonomy, AnalysisAgent performs open-ended risk reasoning over the user query Q and the agent’s intended action trajectory τ , dynamically interacting with a lifelong risk knowledge library $\mathcal{K}_{\mathcal{R}}$ to determine whether the current context triggers a known risk or represents an emerging threat that must be inducted as a new category.

Risk Analysis. Along with the environmental context $\mathcal{C}$, AnalysisAgent conducts joint semantic understanding of the query and actions to identify the risk category $\mathcal{R}$ and determine whether guardrail tool activation is necessary ($z \in \{0, 1\}$):

$$(\mathcal{R}, z) = f_{\text{analysis}}(Q, \tau, \mathcal{C}, \mathcal{K}_{\mathcal{R}}). \quad (5)$$

Tool Generation. When intervention is required ($z = 1$), AnalysisAgent generates a targeted, executable Python guardrail tool $\mathcal{G}$ tailored to the specific risk scenario:

$$\mathcal{G} = f_{\text{gen}}(\mathcal{Q}, \tau, \mathcal{C}, \mathcal{R}). \quad (6)$$

This on-demand invocation mechanism parallels the human judgment of “is this situation dangerous enough to warrant intervention?”—avoiding over-reliance on protective tools when the context is benign, while ensuring rapid response when genuine threats emerge. Just as a human expert codifies a newly discovered vulnerability into a reproducible detection script rather than an informal verbal warning, EVOLVE externalizes each risk judgment into a precise, portable, and reusable code artifact. This ensures that the same protective logic can transfer across different backbone models and application domains without redesign.

SimulationAgent (Optional). In deployment scenarios where only the user instruction $\mathcal{Q}$ is available without an explicit action sequence, SimulationAgent bridges this gap by inferring the most likely action chain $\tau = f_{\text{sim}}(\mathcal{Q}, \mathcal{C})$, analogous to how a security reviewer mentally simulates what a system would do before assessing its safety.

3.6 Reuse: Tool Retrieval and Evolutionary Fusion

Human expertise accumulates not by solving every problem from scratch, but by recognizing structural similarities between new challenges and past solutions, then adapting existing knowledge to fit the new context [15]. FusionAgent instantiates this principle of cumulative cultural evolution for guardrail tools. Rather than allowing the tool library to grow unboundedly with redundant entries, FusionAgent actively searches for semantically related existing tools and fuses them with newly generated ones, enabling each tool to become progressively more capable through iterative refinement.

We define the lifelong guardrail tool library $\mathcal{K}_{\mathcal{G}} = \{(T_i, R_i, \phi_i)\}_{i=1}^N$, where T_i is an audited guardrail tool, R_i its target risk category, and ϕ_i its functional description.

Hierarchical Retrieval. FusionAgent employs a two-stage retrieval mechanism that mirrors how experts first narrow the search space by domain, then perform deep comparison. The first stage filters by risk category:

$$\mathcal{M}_{\mathcal{R}} = \{T_i \mid R_i = \mathcal{R}\}. \quad (7)$$

The second stage uses the language model to evaluate functional-logic alignment between candidates and the new tool, outputting a match indicator $m \in \{0, 1\}$ and the most similar tool T_{sim} :

$$(m, T_{\text{sim}}) = f_{\text{match}}(\mathcal{M}_{\mathcal{R}}, \mathcal{G}). \quad (8)$$

Tool Fusion. When a reusable tool exists ($m = 1$), FusionAgent integrates the new tool’s logic into the existing one, producing an evolved tool $\mathcal{G}'$ with broader coverage:

$$\mathcal{G}' = f_{\text{fuse}}(\mathcal{G}, T_{\text{sim}}, \mathcal{R}). \quad (9)$$

This fusion process is the mechanism through which tools self-evolve: each reuse cycle deepens and extends existing tools rather than leaving them static. When no match is found ($m = 0$), the new tool is retained as a candidate for the library. In both cases, downstream auditing must be passed before library admission, ensuring quality control over the evolutionary process.

3.7 Audit: Sandboxed Execution and Independent Verification

In human governance, trust is not granted by default but earned through independent oversight. Financial systems require external auditors; pharmaceutical development mandates independent clinical trials; legal systems separate prosecution from adjudication [17]. EVOLVE adopts the same principle: tools produced by the generation and reuse stages are not blindly trusted, but must survive independent execution and security review before deployment [40, 41]. This stage is realized through two tightly coupled components: `ExecutorAgent` provides the evidence, and `AuditorAgent` delivers the verdict.

ExecutorAgent: Evidence Production. `ExecutorAgent` operates within a strictly isolated sandbox environment, executing both the guardrail tool and the agent’s intended actions to produce verifiable execution evidence:

$$(\mathcal{E}_{\mathcal{G}'}, \mathcal{E}_{\tau}) = f_{\text{exec}}(\mathcal{Q}, \tau, \mathcal{C}, \mathcal{G}'). \quad (10)$$

The sandbox isolation ensures that neither guardrail tools nor protected operations produce irreversible side effects on the real environment, drawing on the LLM-simulated sandbox paradigm [50]. In practical deployments, Docker-based real execution environments are also supported for higher fidelity requirements.

AuditorAgent: Independent Security Audit. `AuditorAgent` reviews the evolved tool $\mathcal{G}'$ and its execution evidence from a vulnerability perspective, checking for ten categories of typical CWE security vulnerabilities (Appendix ??), producing an audit verdict $D \in \{0, 1\}$ together with a detailed vulnerability report $\mathcal{V}$ describing specific findings. This independence by design ensures that even if upstream modules introduce defects during tool generation or optimization, the audit mechanism can intercept unsafe tools before they enter the library—suppressing the risk of misevolution caused by the Vulnerable Guardrail Tool (VGT) phenomenon we identify in this work.

Final Security Decision. Beyond tool auditing, `AuditorAgent` synthesizes full-pipeline signals—the guardrail tool execution result $\mathcal{E}_{\mathcal{G}'}$, the simulated agent action result $\mathcal{E}_{\tau}$, and the tool’s logical integrity—to deliver the final security verdict y :

$$y = f_{\text{audit}}(\mathcal{Q}, \tau, \mathcal{G}', \mathcal{E}_{\mathcal{G}'}, \mathcal{E}_{\tau}). \quad (11)$$

3.8 Refinement: Iterative Repair and Adaptive Replanning

Resilient systems do not merely detect failures, instead they learn from them. In engineering practice, a failed safety inspection triggers root-cause analysis and targeted repair rather than wholesale rejection [18]. EVOLVE’s refinement stage embodies this corrective learning loop at two levels: repairing flawed tools and replanning unsafe actions.

Tool Repair. When a tool fails the audit ($D = 0$), AuditorAgent performs targeted repair guided by the vulnerability report $\mathcal{V}$, generating a repaired version $\mathcal{G}''$:

$$\mathcal{G}'' = f_{\text{repair}}(\mathcal{G}', \mathcal{V}). \quad (12)$$

This audit-repair cycle executes at most three rounds; tools that still fail are discarded. Only tools passing the audit are admitted to the lifelong tool library:

$$\mathcal{K}_{\mathcal{G}} \leftarrow \mathcal{K}_{\mathcal{G}} \cup \{\mathcal{G}' \mid D = 1\} \cup \{\mathcal{G}'' \mid D = 1\}. \quad (13)$$

This strict admission control ensures that the tool library’s quality improves monotonically as each admitted tool has passed independent audit, preventing the accumulation of flawed artifacts that could compromise future defense decisions.

RevisionAgent: Intent-Preserving Replanning (Optional). When the final verdict is unsafe ($y = \text{unsafe}$), RevisionAgent extends the framework from binary rejection to proactive risk mitigation. Drawing on the identified risk category $\mathcal{R}$ and the agent’s action space, it replans the action trajectory to preserve the user’s core intent while eliminating unsafe operations:

$$\tilde{\tau} = f_{\text{revise}}(\mathcal{Q}, \tau, \mathcal{C}, \mathcal{R}). \quad (14)$$

This mirrors how a human safety officer does not simply shut down a production line upon discovering a hazard, but redesigns the workflow to achieve the same goal safely. Experimental results in embodied environments validate this capability (Fig. 5): rewriting “place the phone in the microwave” as “place the phone on the table” eliminates the fire hazard while fulfilling the user’s core intent.

In summary, EVOLVE’s four-stage pipeline—generation, reuse, audit, and refinement—forms a continuously verified evolutionary loop. Each stage maps to a well-established principle of human adaptive systems: contextual threat reasoning, cumulative knowledge reuse, independent oversight, and corrective learning. Together, they enable defense capabilities that grow stronger through interaction rather than degrading under scenario shift.

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4 Data Availability

All data needed to evaluate the conclusions is available in <https://github.com/simonsshoot/EVOLVE/>.

5 Code Availability

Full source codes needed to evaluate the conclusions is available in <https://github.com/simonsshoot/EVOLVE/>.

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Appendix A Related Work

Self-evolving agents.

Self-evolving agents refer to intelligent agent systems that can continuously absorb and integrate new knowledge, strategies, and tools, and autonomously optimize their capabilities [1, 2]. There are currently four main directions of self-evolution [40]. The first focuses on model evolution, where agents further refine their parameters through self-generated data, combined with methods such as reinforcement learning [51–53]. The second focuses on memory evolution, where agents continuously store experience during interactions and use past experience to guide future interaction [54–56]. The third focuses on tool evolution, where agents can create, reuse, and optimize tools during interactions, expanding their capability boundaries and improving their proficiency in using tools [57, 58]. The fourth focuses on workflow evolution, where agents can also improve performance by optimizing execution pipelines or collaborative processes [59, 60].

Agent safety.

Given that the risks posed by LLM agents are not limited to natural language output, the methods for protecting against them differ fundamentally from those for LLM guardrail [2, 61]. Most existing LLM agent protection methods lack generalization ability, TrustAgent [62] requires predefined rules, GuardAgent [11] and Consecra [63] rely on manually specified trusted contexts. Although some works have begun to focus on adaptation [28, 52], it is still insufficient for LLM agents with diverse output modes and security requirements. Our EVOLVE framework, which empowers agents to evolve on their own, will be key to bridging the problem.

Appendix B Prompt Correctness Validation

Since most components of EVOLVE are implemented as LLM API calls with task-specific prompts, we conduct experiments to validate that our prompts elicit correct and reliable outputs. We divide the modules into two categories based on their role: *non-decision modules* (whose outputs are intermediate representations such as risk analyses, generated tools, and action simulations) and *decision modules* (whose outputs directly determine the safe/unsafe verdict). We validate each category with a different methodology.

❶ **Non-Decision Modules: Multi-Model Voting.** Non-decision modules include SimulationAgent (f_{sim}), AnalysisAgent’s risk analysis (f_{analysis}), AnalysisAgent’s tool generation (f_{gen}), and FusionAgent ($f_{\text{match}}, f_{\text{fuse}}$). Since there is no single ground-truth answer for these open-ended generation tasks, we adopt a multi-model consensus approach to validate output quality. For each module, we randomly sample 50 inputs from the RJudge dataset and collect outputs from all four backbone models used in our experiments (GPT-4o, DeepSeek-V3, Gemini-2.5-pro, and Claude-sonnet-4-5). For each model, a human annotator extracts its key information points and compares them against the consensus points from the other three models. The coverage rate measures the proportion of consensus points that each model’s output

successfully covers. Across all four models and all non-decision modules, the coverage rate consistently exceeds 90%, indicating that our prompts reliably elicit outputs capturing the essential information regardless of the backbone model.

② **Decision Modules: Agreement with Human Judgment.** Decision modules include AuditorAgent’s tool audit (f_{audit}) and the final safety verdict (DOUBT_DECISION_PROMPT). These modules produce binary safe/unsafe decisions that directly affect the framework’s outcome. We randomly sample 150 cases from across the three benchmarks (balanced between safe and unsafe) and have a human expert independently provide ground-truth decisions, experiments are done on four backbone LLMs. The agreement rate between the LLM decisions produced by our prompts and the human expert judgments constantly reaches 100%, confirming that our decision prompts faithfully align with human safety reasoning.

Appendix C Main Results

Table C1 reports the complete numerical results across three benchmarks and four backbone LLMs. EVOLVE achieves the highest F1 score in all 12 benchmark–model combinations, with average improvements of +9.0% on RJudge, +7.4% on ASSEBench, and +41.1% on AgentHarm over the respective runner-up methods. Notably, EVOLVE maintains F1 scores stably between 0.86 and 0.92 regardless of the backbone choice, while baseline methods exhibit cross-benchmark fluctuations of up to 65% (ShieldAgent on AgentHarm). ShieldAgent is a training-based method that does not rely on a backbone LLM; its results are identical across all LLM groups.

Table C1: Results across three benchmarks and four backbone LLMs. **Bold** indicates the best, Underline indicates the second. [†]ShieldAgent does not rely on a backbone LLM, its scores are identical across all groups.

Method	RJudge				ASSEBench				AgentHarm			
	Acc	Prec	Rec	F1	Acc	Prec	Rec	F1	Acc	Prec	Rec	F1
<i>Backbone LLM: DeepSeek</i>												
CoTSafe	<u>0.8091</u>	<u>0.8575</u>	0.7223	0.7815	0.6316	0.7629	<u>0.8578</u>	0.7722	0.7666	<u>0.9007</u>	0.5978	0.6915
ShieldAgent [†]	0.7863	0.7136	0.9934	<u>0.8318</u>	0.8046	0.8762	0.7532	0.8095	0.5286	0.6923	0.1029	0.1791
AGrail	0.5403	0.6488	<u>0.9168</u>	0.7601	<u>0.8678</u>	0.9213	0.8039	<u>0.8587</u>	0.6000	0.8214	0.3557	0.4964
GuardAgent	0.6935	0.7684	0.5037	0.6086	0.7259	0.6294	0.9240	0.7485	<u>0.7714</u>	0.8992	<u>0.6114</u>	<u>0.7281</u>
EVOLVE	0.9107	0.8982	0.9148	0.9065	0.8868	<u>0.8974</u>	0.8430	0.8690	0.8800	0.9346	0.8171	0.8719
<i>Backbone LLM: GPT-4o</i>												
CoTSafe	0.6786	0.7560	0.4704	0.5804	0.4663	0.4776	0.7328	0.5781	0.5857	0.6442	<u>0.3829</u>	<u>0.4801</u>
ShieldAgent [†]	<u>0.7863</u>	0.7136	0.9934	<u>0.8318</u>	<u>0.8046</u>	0.8762	<u>0.7532</u>	<u>0.8095</u>	0.5286	0.6923	0.1029	0.1791
AGrail	0.5494	0.4889	0.4933	0.4910	0.6924	<u>0.9120</u>	0.6344	0.7482	<u>0.5874</u>	1.0000	0.1771	0.3009
GuardAgent	0.6848	<u>0.8571</u>	0.4000	0.5455	0.6184	0.6086	0.6609	0.6336	0.5829	<u>0.8222</u>	0.2114	0.3362
EVOLVE	0.9002	0.9338	<u>0.8916</u>	0.9123	0.8633	0.9368	0.8420	0.8869	0.8758	1.0000	0.8240	0.9035
<i>Backbone LLM: Claude</i>												
CoTSafe	<u>0.8634</u>	<u>0.9288</u>	0.7516	0.8309	0.7466	0.6967	0.8725	0.7746	0.6543	1.0000	0.3086	0.4718
ShieldAgent [†]	0.7863	0.7136	0.9934	<u>0.8318</u>	0.8046	<u>0.8762</u>	0.7532	<u>0.8095</u>	0.5286	0.6923	0.1029	0.1791
AGrail	0.7916	0.7436	<u>0.8625</u>	0.7987	<u>0.8152</u>	0.7037	<u>0.9209</u>	0.7981	0.5543	0.6479	0.2629	0.3741
GuardAgent	0.6856	0.7018	0.4513	0.5489	0.7143	0.6529	0.9314	0.7675	<u>0.7600</u>	0.8897	<u>0.6437</u>	<u>0.7467</u>
EVOLVE	0.8841	0.9349	0.8385	0.8841	0.8802	0.9000	0.8529	0.8758	0.8812	<u>0.9854</u>	0.8272	0.8994
<i>Backbone LLM: Gemini</i>												
CoTSafe	<u>0.8021</u>	0.8520	0.7037	0.7703	0.6242	0.4863	0.8211	0.6107	0.7771	0.8392	<u>0.6857</u>	<u>0.7547</u>
ShieldAgent [†]	0.7863	0.7136	0.9934	<u>0.8318</u>	0.8046	0.8762	0.7532	0.8095	0.5286	0.6923	0.1029	0.1791
AGrail	0.7846	<u>0.9077</u>	0.5801	0.7072	<u>0.8849</u>	<u>0.9062</u>	0.8352	<u>0.8692</u>	0.5457	<u>0.9000</u>	0.1029	0.1846
GuardAgent	0.6968	0.7904	0.4889	0.6042	0.7234	0.6585	0.9265	0.7684	<u>0.8333</u>	<u>0.9000</u>	0.6171	0.7297
EVOLVE	0.9255	0.9257	<u>0.9222</u>	0.9240	0.8874	0.9136	<u>0.8554</u>	0.8834	0.8886	0.9851	0.8468	0.9107